GATE ME 90 Days Challenge

Day 35/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Thermodynamics	Basics & Pure Substances	71 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on thermodynamic systems and properties, zeroth law and temperature scales, displacement work along different paths, phase change of water, steam quality from property tables, the T–v diagram, ideal-gas equation, compressibility factor and the van der Waals gas. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A pressure cooker is on the stove; steam escapes intermittently through the weight valve while the food cooks. Taking the cooker and its contents as the system, it is

(A) a closed system
(B) an open system
(C) an isolated system
(D) an adiabatic closed system

Q2 NAT

A thermocouple used as a thermometer gives an emf $e = 0.2t - 5 \times 10^{-4} t^2$ (mV), where t is in °C. It is calibrated to read 0 at the ice point and 100 at the steam point, its scale being linear in e. When the gas thermometer reads 50 °C, the thermocouple thermometer reads (round off to two decimal places)

_____

Q3 MCQ

Which statement about a quasi-static process is NOT correct?

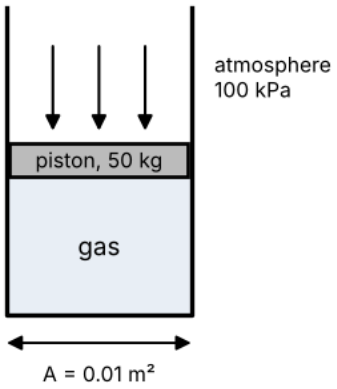
(A) It is a succession of equilibrium states
(B) It can be shown as a continuous line on a property diagram
(C) Free expansion of a gas into vacuum is a quasi-static process
(D) Work in a quasi-static compression of a gas is $\int p\, dV$

Q4 NAT

A gas expands quasi-statically from 0.1 m^3 to 0.3 m^3 following the law $p = c/V^2$, the initial pressure being 600 kPa. The work done by the gas (in kJ) is _____

Q5 NAT

A gas is held in a vertical cylinder of cross-section 0.01 m² by a frictionless piston of mass 50 kg, as shown; the atmosphere above is at 100 kPa. The gas is heated slowly and its volume increases by 0.002 m³. Take g = 9.81 m/s². The work done by the gas (in J, round off to one decimal place) is _____



Q6 MCQ

Wet steam at 1 MPa has a specific enthalpy of 2000 kJ/kg. At 1 MPa, h_f = 762.8 kJ/kg and h_{fg} = 2015.3 kJ/kg. Its dryness fraction is about

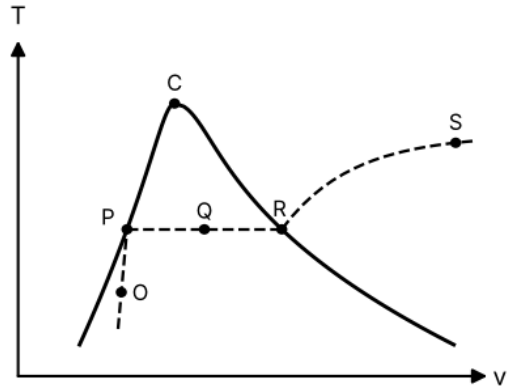
(A) 0.992
(B) 0.381
(C) 0.614
(D) 0.719

Q7 NAT

Wet steam at 200 kPa has a dryness fraction of 0.85. At 200 kPa, v_f = 0.001061 m³/kg and v_g = 0.8857 m³/kg. Its specific volume (in m³/kg, round off to three decimal places) is _____

Q8 MCQ

The T–v diagram of water shows the saturation dome with critical point C and one constant-pressure line through the points O, P, Q, R and S. Which statement is correct?



(A) O is compressed liquid, Q is a wet mixture, S is superheated vapour
(B) O is a wet mixture, R is saturated liquid, S is compressed liquid
(C) P is saturated vapour and R is saturated liquid
(D) Q is superheated vapour because its temperature equals that at R

Q9 MCQ

When ice is heated slowly at a constant pressure BELOW the triple-point pressure, it

(A) melts first and then evaporates
(B) melts and the water then freezes again
(C) remains solid at all temperatures
(D) sublimes directly into vapour

Q10 NAT

0.5 kg of an ideal gas occupies 0.2 m^3 at 300 kPa and 87 °C. Taking the universal gas constant as 8.314 kJ/kmol·K, the molar mass of the gas (in kg/kmol, round off to two decimal places) is _____

Q11 MCQ

In the van der Waals equation $(p + a/v^2)(v - b) = RT$, the constants a and b account respectively for

(A) the finite volume of the molecules and intermolecular attraction
(B) intermolecular attraction and the finite volume of the molecules
(C) molecular collisions with the wall and molecular vibration
(D) the compressibility of the gas and its specific heat

Q12 NAT

Nitrogen (R = 0.2968 kJ/kg·K) at 10 MPa and 150 K has a measured specific volume of 0.00375 m^3/kg. Its compressibility factor (round off to three decimal places) is _____

Q13 MCQ

An ideal gas in a sealed rigid container at 200 kPa is heated from 27 °C to 127 °C. Its final pressure is

(A) 940.7 kPa
(B) 150.0 kPa
(C) 254.0 kPa
(D) 266.7 kPa

Q14 NAT

A gas expands quasi-statically and polytropically (pV^n = constant) from 1000 kPa, 0.1 m^3 to 200 kPa, 0.4 m^3. The work done by the gas (in kJ, round off to one decimal place) is _____

Q15 MCQ

A rigid, insulated tank is divided into two equal parts by a partition. One part contains an ideal gas; the other is evacuated. The partition is removed. For the gas, the work done W and the final temperature T_2 (initial T_1) are

(A) $W > 0, T_2 < T_1$
(B) $W = 0, T_2 < T_1$
(C) $W = 0, T_2 = T_1$
(D) $W = p_1V_1 \ln 2, T_2 = T_1$

Answer Key

Q	Type	Answer
Q1	MCQ	(B) an open system
Q2	NAT	58.33 (accept 58.0 to 58.7)
Q3	MCQ	(C) Free expansion of a gas into vacuum is a quasi-static process
Q4	NAT	40
Q5	NAT	298.1 (accept 296.6 to 299.6)
Q6	MCQ	(C) 0.614
Q7	NAT	0.753 (accept 0.749 to 0.757)
Q8	MCQ	(A) O is compressed liquid, Q is a wet mixture, S is superheated vapour
Q9	MCQ	(D) sublimes directly into vapour
Q10	NAT	24.94 (accept 24.8 to 25.1)
Q11	MCQ	(B) intermolecular attraction and the finite volume of the molecules
Q12	NAT	0.842 (accept 0.838 to 0.846)
Q13	MCQ	(D) 266.7 kPa
Q14	NAT	124.3 (accept 123.0 to 125.5)
Q15	MCQ	(C) $W = 0$, $T_2 = T_1$

Detailed Solutions

Q1 **Answer: (B) an open system**

Concept: A system is open (a control volume) if **mass** crosses its boundary; closed if only energy crosses; isolated if neither does.

Heat enters from the stove and steam (mass) leaves through the valve, so mass crosses the boundary: an **open system**.

Exam tip: A sealed rigid tank being heated is closed; a thermos flask (ideally) is isolated; turbines, nozzles, pumps and compressors are open systems.

Q2 **Answer: 58.33 (accept 58.0 to 58.7)**

Concept: A thermometer scale linear in its thermometric property: $t^* = 100 \times (e - e_{ice})/(e_{steam} - e_{ice})$.

$e(0) = 0$; $e(100) = 20 - 5 = 15$ mV; $e(50) = 10 - 1.25 = 8.75$ mV

$t^* = 100 \times 8.75/15 =$ **58.33**

Exam tip: Two thermometers agree only at their fixed points; between them they differ when the property is not linear in t. The ideal-gas scale is the reference.

Q3 Answer: (C) Free expansion of a gas into vacuum is a quasi-static process

Concept: A quasi-static process is carried out infinitely slowly, so the system passes through equilibrium states only.

Free (unresisted) expansion is sudden and far from equilibrium: intermediate states are undefined and cannot be drawn. So (C) is wrong.

Exam tip: $\int p\,dV$ gives the work only for a quasi-static process; in free expansion $\int p\,dV \neq 0$ but the actual work is zero.

Q4 Answer: 40

Concept: $W = \int p\,dV$ along the given path.

$c = p_1V_1^2 = 600 \times 0.1^2 = 6$ kPa·m^6

$W = \int c/V^2\,dV = c\,(1/V_1 - 1/V_2) = 6 \times (10 - 3.333) =$ **40 kJ**

Exam tip: Do not use the polytropic formula blindly; here n = 2, and $(p_1V_1 - p_2V_2)/(n - 1) = (60 - 20)/1 = 40$ kJ gives the same result as a check.

Q5 Answer: 298.1 (accept 296.6 to 299.6)

Concept: A free piston keeps the gas pressure constant at $p = p_{atm} + mg/A$, so $W = p\Delta V$.

$p = 100 + (50 \times 9.81/0.01)/1000 = 100 + 49.05 = 149.05$ kPa

$W = 149.05 \times 0.002$ kJ $= 0.2981$ kJ $=$ **298.1 J**

Of this, 200 J pushes back the atmosphere and 98.1 J lifts the piston ($mg\Delta h = 50 \times 9.81 \times 0.2$).

Exam tip: Forgetting p_{atm} (using only mg/A) is the most common mistake.

Q6 Answer: (C) 0.614

Concept: For a wet mixture $h = h_f + x\,h_{fg}$.

$x = (2000 - 762.8)/2015.3 = 1237.2/2015.3 =$ **0.614**

Exam tip: 0.992 comes from 2000/2015.3, i.e. forgetting h_f. Remember h_{fg} (not h_g) multiplies x.

Q7 Answer: 0.753 (accept 0.749 to 0.757)

Concept: $v = v_f + x(v_g - v_f) \approx x\,v_g$ at low pressure.

$v = 0.001061 + 0.85 \times (0.8857 - 0.001061) =$ **0.753 m³/kg**

Approximation $x\,v_g = 0.7528$, almost the same because $v_f \ll v_g$.

Exam tip: The approximation fails near the critical point, where v_f and v_g are comparable.

Q8 Answer: (A) O is compressed liquid, Q is a wet mixture, S is superheated vapour

Concept: Left of the saturated-liquid line: compressed (subcooled) liquid; under the dome: liquid–vapour mixture; right of the saturated-vapour line: superheated vapour.

O lies left of the dome (compressed liquid); P is saturated liquid; Q is a wet mixture; R is saturated vapour; S is superheated.

Along P–Q–R both T and p stay constant (phase change), so the isobar is horizontal inside the dome.

Exam tip: Inside the dome p and T are not independent; you need x (or v, h, s) to fix the state.

Q9 **Answer: (D) sublimes directly into vapour**

Concept: On the p–T phase diagram the liquid phase exists only above the triple-point pressure (0.6117 kPa for water).

Below this pressure, the constant-pressure line crosses only the sublimation curve: solid → vapour directly.

Exam tip: Water's fusion line slopes backwards (ice is less dense than water), so raising the pressure lowers the melting point; CO_2 sublimes at 1 atm because its triple point is at 5.1 atm.

Q10 **Answer: 24.94 (accept 24.8 to 25.1)**

Concept: pV = mRT gives R; then M = $\bar{R}$/R.

R = pV/(mT) = 300 × 0.2/(0.5 × 360) = 0.3333 kJ/kg·K (T = 87 + 273 = 360 K)

M = 8.314/0.3333 = **24.94 kg/kmol**

Exam tip: Using 87 instead of 360 K gives an absurd molar mass of about 6.0; always convert to kelvin.

Q11 **Answer: (B) intermolecular attraction and the finite volume of the molecules**

Concept: a/v^2 is a pressure correction: attraction between molecules reduces the pressure they exert on the wall, so it is added to p.

b is the co-volume: the space taken up by the molecules themselves is subtracted from v.

Exam tip: From the critical-point conditions $(\partial p/\partial v)_T = 0$ and $(\partial^2 p/\partial v^2)_T = 0$: $a = 27R^2T_c^2/(64p_c)$, $b = RT_c/(8p_c)$.

Q12 **Answer: 0.842 (accept 0.838 to 0.846)**

Concept: Z = pv/(RT) compares the real volume with the ideal-gas volume.

Z = 10000 × 0.00375/(0.2968 × 150) = 37.5/44.52 = **0.842**

Z < 1: attractive forces dominate at this state (close to the critical point of N_2: 126 K, 3.4 MPa).

Exam tip: With p in kPa, R in kJ/kg·K and v in m^3/kg, the units cancel directly.

Q13 **Answer: (D) 266.7 kPa**

Concept: Rigid and sealed: m and V constant, so p/T = constant (absolute T).

p_2 = 200 × (127 + 273)/(27 + 273) = 200 × 400/300 = **266.7 kPa**

Exam tip: 940.7 kPa comes from using Celsius temperatures (200 × 127/27). Absolute pressure and absolute temperature, always.

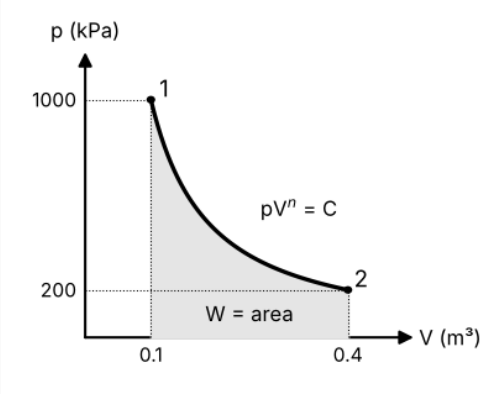
Q14 **Answer: 124.3 (accept 123.0 to 125.5)**

Concept: Find n from the end states, then $W = (p_1V_1 - p_2V_2)/(n - 1)$.

$n = \ln(p_1/p_2)/\ln(V_2/V_1) = \ln 5/\ln 4 = 1.161$

$W = (1000 \times 0.1 - 200 \times 0.4)/(1.161 - 1) = 20/0.161 =$ **124.3 kJ** (the shaded area in the figure)

Exam tip: The result is very sensitive to n; keep at least three decimals of n (rounding n to 1.16 gives 125.0 kJ).



Q15 **Answer: (C) $W = 0, T_2 = T_1$**

Concept: In free expansion against vacuum there is no resistance, so no work is done; the tank is insulated, so Q = 0.

First law: $\Delta U = Q - W = 0$. For an ideal gas U depends only on T, so $T_2 = T_1$ (the pressure halves).

Exam tip: $p_1V_1 \ln 2$ is the work of a reversible isothermal expansion between the same states; free expansion is irreversible and does no work.

What next?

Day 35 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 36/90: Weekly Revision, Week 6 Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.